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ECE 4644 / 5610 Satellite Communications

Homework Assignment #9 Solutions

Problem #1

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BPSK and QPSK modulation encode information using 1-bit and 2-bit symbols, respectively. Let's propose an OPSK (Octa-) modulation that uses 3-bit symbols.

a) Provide a truth table that associates 3-bit words arranged in sequential binary order (000 001 010 etc.) with sequential phase angles and sketch as a constellation diagram.

b) Consider transitions between each phase angle and its adjacent phase angles. How many 1-bit, 2-bit, and 3-bit transitions are possible?

c) Explain the defining characteristic of a Gray code. Now derive a mapping scheme for a 3-bit Gray code.

d) Indicate the Gray code values on the constellation diagram of part (a). How many 1-bit transitions to adjacent phase angles are possible?

e) What do you expect will be the main advantage and main disadvantage of OPSK modulation compared to BPSK? Try to be quantitative to illustrate the meaning of statements that involve 'more' or 'less'.

Problem #2

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A QPSK satellite link operates with $\alpha = 0.25$ RRC filters and a bit rate of 2.0 Mbps. The overall C/N in clear air is 15 dB and the implementation margin is 1.0 dB.

a) Find the noise bandwidth of the receiver and the occupied bandwidth of the RF signal.

b) Find the BER in clear air and the average time between bit errors at the receiver output. (Hint: apply the BER and bit rate to determine the number of bit errors per second, then invert.)

c) Rain affects the uplink and reduces the C/N there by 3 dB. The transponder is linear and the downlink is free of rain. Find the new overall C/N, the BER, and the average time between bit errors at the receiver output. Comment.

Problem #3

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Consider the design of a (4, 2) block code to perform ½ FEC on 2-bit messages carried over a transmission link.

a) List all 16 possible codewords.

b) Construct a truth table that maps all the possible 2-bit messages into a subset of the codewords with a minimum distance of 2.

c) If any of the other 12 codewords appears at the receiver output an error has occurred. By examining the distance of each codeword from the accepted codewords and assuming only 1-bit errors, determine which codewords can be corrected and state their corrections. Construct a Table that summarizes over all 16 codewords that could come through the link. (This Table will tell you want to make of each codeword.)

Problem #4

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In a short paragraph explain what is meant by FEC (why *forward*?) and describe three techniques for correcting bit errors with FEC (2-3 sentences per technique).

Problem #5

Five earth stations share one transponder of a 6/4 GHz satellite. The stations all operate in TDMA mode. Speech signals are sampled at 8 kHz using 8 bits/sample. The sampled signals (PCM) are then multiplexed into 2.0 Mbps streams at each station using QPSK without FEC.

a) Find the bit rate for each PCM signal.

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b) Find the number of speech signals (as PCM) that can be sent by each earth station, as a single access, with no overhead (i.e., no header or CRC) or guard times. This is the TDM data stream.

c) State the shortest frame time for any TDMA scheme.

d) Find the number of speech signals (as PCM) that can be sent by each earth station, as a single access, with a 1 ms frame, 10 μs guard time and a 10 μs header per frame.

$\sum = 31$

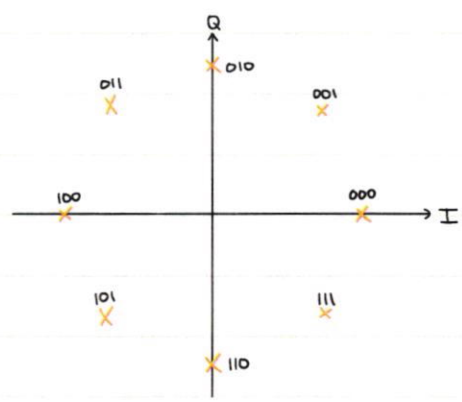
Problem 1

BPSK and QPSK modulation encode information using 1-bit and 2-bit symbols, respectively. Let's propose an OPSK modulation that uses 3-bit symbols

a) Provide a truth table that associates 3-bit words arranged in sequential binary order with sequential phase angles and sketch as a constellation diagram

Word	Phase [deg]
000	0
001	45°
010	90°
011	135°
100	180°
101	225°
110	270°
111	315°

Constellation:



Each word is 3 bits, so there are $2^3 = 8$ possible words. Equally spaced around a circle, each word will be $360°/8 = 45°$ apart. When assigned in simple binary order, the table and constellation above is achieved.

b) Consider transitions between each phase angle and its adjacent phase angles. How many 1-bit, 2-bit, and 3-bit transitions are possible?

Given 8 states and two possible transitions per state (previous phase, next phase), there are $8 \times 2 = 16$ transitions to observe

Phase			Word			Bits Changed	
Current	Previous	Next	Current	Previous	Next	Previous	Next
0	315°	45°	000	111	001	3	1
45°	0°	90°	001	000	010	1	2
90°	45°	135°	010	001	011	2	1
135°	90°	180°	011	010	100	1	3
180°	135°	225°	100	011	101	3	1
225°	180°	270°	101	100	110	1	2
270°	225°	315°	110	101	111	2	1
315°	270°	0°	111	110	000	1	3

number of: 1-bit transitions = 8

2-bit transitions = 4

3-bit transitions = 4

✓

c) Explain the defining characteristic of a Gray code. Now derive a mapping scheme for a 3-bit Gray code.

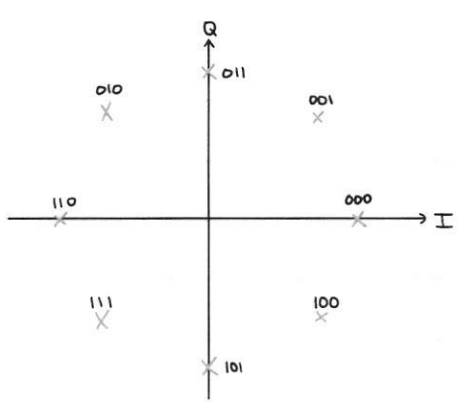
A Gray code is a mapping of binary words to states such that adjacent states only differ by one bit. Adjacent state errors, which are the most common, will produce an output that is only incorrect by one bit. This makes error detection and correction much easier.

Original	Gray
000	000
001	001
010	011
011	010
100	110
101	111
110	101
111	100

Each word in the Gray code is different from the previous word by only one bit.

d) Indicate the Gray code values on the constellation diagram of part (a). How many 1-bit transitions to adjacent phase angles are possible?

Constellation:



There are 8 states and each state has 2 transitions, so there are 16 total adjacent transitions. Given the Gray code ordering, each transition only changes 1 bit. Therefore, there are 16 1-bit transitions.

e) What do you expect will be the main advantage and main disadvantage of OPSK modulation compared to BPSK? Try to be quantitative to illustrate the meaning of statements that involve "more" or "less"

The main advantage of OPSK over BPSK is higher bit rate. Each BPSK symbol only carries 1 bit of information, while each OPSK symbol carries 3 bits of information. Assuming equal symbol rates, OPSK has 3x the bit rate of BPSK. The main disadvantage of OPSK is its bit error rate (BER). For BPSK, adjacent states are 180° apart, so the recieved phase must be incorrect by at least 90° to cause a bit error. With OPSK, adjacent states are just 45° apart, so a phase error of just 22.5° will cause a bit error. Thus, OPSK is much more susceptible to noise than BPSK. To maintain equivalent BER, OPSK requires 6 dB higher C/N compared to BPSK

why? explain / justify

- while a complete analysis would require a statistical ~~analysis~~ treatment, it can be inferred from BPSK → QPSK requiring 3 dB more C/N to equalize BER that BPSK → OPSK would require ~ 6 dB more C/N

Problem 2

A QPSK satellite link operates with $\alpha = 0.25$ RRC filters and a bit rate of 2.0 Mbps. The overall C/N in clear air is 15 dB and the implementation margin is 1.0 dB.

a) Find the noise bandwidth of the receiver and the occupied bandwidth of the RF signal.

RF Link:

$$R_s = \underbrace{R_b/2}_{\text{for BPSK}} = \frac{2.0\text{ Mbps}}{2} = 1.0\text{ Msps}$$

$$B_n = R_s \rightarrow B_n = 1\text{ MHz}$$

$$B_{occ} = R_s(1+\alpha) \rightarrow B_{occ} = 1\text{M}(1+0.25) = 1.25\text{ MHz}$$

b) Find the BER in clear air and the average time between bit errors at the receiver output.

$$(C/N)_{eff} = (C/N)_{clear} - IM$$

$$(C/N)_{eff} = 15\text{ dB} - 1.0\text{ dB} = 14\text{ dB}$$

$$14\text{ dB} \xrightarrow{\text{linear}} 10^{14/10} = 25.1$$

$$BER = Q\left(\sqrt{C/N}\right) = Q(5.01)$$

$$BER = 2.872\text{ E-}7$$

From Appendix C, Q(z) for z=5.0

Time between errors:

$$BER = 2.827\times10^{-7}\ \frac{\text{errors}}{\text{bit}}$$

$$R_b = 2.0\text{ M}\ \frac{\text{bits}}{\text{s}}$$

$$BER\cdot R_b \left[\frac{\text{erros}}{\text{bit}}\cdot\frac{\text{bits}}{\text{s}}\right] = 2.827\times10^{-7}\cdot 2.0\times10^{6}\left[\frac{\text{errors}}{\text{s}}\right]$$

$$= 0.5654\ \frac{\text{errors}}{\text{s}}$$

OR, inverted $\frac{1}{0.5654}\left[\frac{\text{s}}{\text{error}}\right]$

1.77 seconds/error

c) Rain affects the uplink and reduces the C/N there by 3 dB. The transponder is linear and the downlink is free of rain. Find the new overall C/N, the BER, and the average time between bit errors at the receiver output.

$(C/N)_{up} \downarrow 3dB = 12\ dB$

$(C/N)_{dn} \downarrow 3dB = 12\ dB$ (because of linear transponder)

$\underline{(C/N)_0 \downarrow 3dB = 12 dB}$

$$(C/N)_{eff\_rain} = (C/N)_{0\_rain} - IM = 12dB - 1\ dB = 11 dB$$

$$BER = Q\left(\sqrt{(C/N)_{linear}}\right) = Q\left(\sqrt{10^{11/10}}\right) = Q\left(\sqrt{12.58}\right) = Q(3.5)$$

$\rightarrow$ Appendix C $\rightarrow$ $\underline{BER = 2.332E\text{-}4}$

$$BER \cdot R_b = 2.332 \times 10^{-4} \cdot 2 \times 10^{6} \left[\frac{errors}{s}\right] = 466 \frac{errors}{s}$$

inverted

$\underline{2.1\ ms/error}$

3 dB rain attenuation on the uplink causes errors to happen ~825x more frequently.

Problem 3

Consider the design of a (4,2) block code to perform ½ FEC on 2-bit messages carried over a transmission link.

(4,2) block code: 2 message bits, 2 parity bits

a) List all 16 possible codewords:

0000	0100	1000	1100
0001	0101	1001	1101
0010	0110	1010	1110
0011	0111	1011	1111

✓

b) Construct a truth table that maps all the possible 2-bit messages into a subset of the codewords with a minimum distance of 2.

Choosing diagonal elements of table in 3a, a set of base codewords are as follows:

Message	Codeword
00	0000
01	0101
10	1010
11	1111

✓

Each of these codewords contains two message bits followed by a set of identical parity bits. Each codeword is different from every other codeword by at least two bits.

c) If any of the other 12 codewords appears at the receiver output an error has occurred. By examining the distance of each codeword from the accepted codewords and assuming only 1-bit errors, determine which codewords can be corrected and state their corrections. Construct a Table that summarizes over all 16 codewords that could come through the link.

(Actual)	Received	Distance to Closest Codeword	Interpreted
X	0000	0	0000
	0001	1	0000 OR 0101
	0010	1	0000 OR 1010
	0011	2	Any
	0100	1	0000 OR 0101
X	0101	0	0101
	0110	2	Any
	0111	1	0101 OR 1111
	1000	1	0000 OR 1010
	1001	2	Any
X	1010	0	1010
	1011	1	1010 OR 1111
	1100	2	Any
	1101	1	0101 OR 1111
	1110	1	1010 OR 1111
X	1111	0	1111

From Week 12 Slides:

maximum bit errors <u>detected</u>

$= d_{min} - 1 = 2-1 = 1$

maximum bit errors <u>corrected</u>

$= \frac{1}{2}(d_{min}-1) = \frac{1}{2}(2-1) = 0.5$

↓

Thus, this (4,2) block code can <u>detect</u> all 1-bit errors but cannot <u>correct</u> any

✓

As shown in the table above, given the (4,2) block code, all 1- and 2-bit errors were detected, but none were correctable. Because the codewords were different by only two bits, all received words with minimum distance of 1 could map to two different codewords. Words of distance 2 were equally far from all valid codewords. ✓

Problem 4

In a short paragraph explain what is meant by FEC (why forward?) and describe three techniques for correcting bit errors with FEC.

FEC, which stands for forward error correction, is a transmission technique which allows the receiver to detect and correct bit errors. It is called "forward" error correction because the receiver does not have to call back to the transmitter for retransmission. Instead, redundant information is sent forward with the original message. FEC is especially useful in satellite communications because propagation delay can be significant, which makes retransmission impractical. A common method of FEC is block coding, where each valid message is paired with extra, redundant bits. These codewords differ by a specified number of bits. If an invalid word is recieved, the message is corrected to be the closest codeword. Another FEC technique is convolutional coding. In this technique, each input bit is turned into several output bits using the current bit and a few previous bits. The received sequence is compared to all possible valid patterns and the most likely one is chosen, so the output can be recovered even if some bits are corrupted. A third FEC technique is interleaving. While it doesn't detect or correct codes on its own, interleaving can be combined with other FEC techniques to improve robustness against burst errors, where several bits are lost in quick succession. By writing data in one order and reading it out in a different order, adjacent transmitted errors are spread out, making it easier for FEC systems to correct.

Problem 5

Five earth stations share one transponder of a 6/4 GHz satellite. The stations all operate in TDMA mode. Speech signals are sampled at 8 kHz using 8 bits/sample. The sampled signals (PCM) are then multiplexed into 2.0 Mbps streams at each station using QPSK without FEC

a) Find the bit rate of each PCM signal

Each signal: $8 \frac{\text{bits}}{\text{sample}} \cdot 8\text{k} \frac{\text{samples}}{\text{sec}} = 64\text{k} \frac{\text{bits}}{\text{s}} = 64 \text{ kbps}$

b) Find the number of speech signals (as PCM) that can be sent by each earth station, as single access, with no overhead or guard times. This is the TDM data stream.

TDMA Frame:

1	2	3	4	5

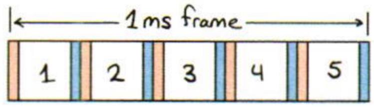
$2.0 \text{ Mbps} / 5 \text{ stations} = 400 \text{ kbps/earth station}$

Each earth station: $\frac{400 \text{ kbps}}{64 \text{ kbps/channel}} = 6.25$ channels $\rightarrow$ 6 voice channels per earth station

c) State the shortest frame time for any TDMA scheme.

Each voice channel is being sampled at 8 kHz. If a frame is sent every time a sample is received, that would happen every $1/8\text{kHz} = 125\ \mu s$. Thus, the minimum practical frame time for this system is $125\ \mu s$. While a shorter frame time is possible, a complete PCM sample would not be available for every frame, so some frames would be sent empty. This is wasteful and impractical, so the shortest practical frame for this system is $125\ \mu s$.

d) Find the number of speech signals (as PCM) that can be sent by each earth station, as a single access, with a 1 ms frame, 10 μs guard time, and a 10 μs header per frame.



10 μs header, 10 μs guard

$$2M \frac{\text{bits}}{s} \cdot 1m \frac{s}{\text{frame}} = 2k \frac{\text{bits}}{\text{frame}}$$

$$2M \frac{\text{bits}}{s} \cdot 10\mu \frac{s}{\text{header}} = 20 \frac{\text{bits}}{\text{header}}$$

$$2M \frac{\text{bits}}{s} \cdot 10\mu \frac{s}{\text{guard}} = 20 \frac{\text{bits}}{\text{guard}}$$

$$2k \frac{\text{bits}}{\text{frame}} \div 5 \text{ stations}$$
$$= 400 \text{ bits/frame/station}$$

✓

Per earth station, in 1 frame:

$$400 \text{ bits} - 20 \frac{\text{bits}}{\text{header}} \cdot 1 \text{ header} - 20 \frac{\text{bits}}{\text{guard}} \cdot 1 \text{guard} = 360 \text{ bits available for Voice channels}$$

Each voice channel: $64k \frac{\text{bits}}{s} \cdot 1m \frac{s}{\text{frame}} = 64 \frac{\text{bits}}{\text{frame}}$

360 bits available $\div$ 64 bits/channel = 5.625 channels

$\rightarrow$ 5 voice channels per station

With a 1 ms frame time, 10 μs header time, 10 μs guard time, total transponder throughput of 2 Mbps, five earth stations, and each voice channel sampled at 8 kHz with 8 bits/sample, each earth station can support up to 5 speech channels. ✓